

UNIVERSITY OF ENERGY AND NATURAL RESOURCES

School of Engineering – Department of Computer Engineering

TITLE

DEVELOPMENT OF A CLINICAL DECISION SUPPORT SYSTEM USING MACHINE LEARNING ALGORITHMS: A COMPARATIVE STUDY ON BREAST CANCER DIAGNOSTIC CLASSIFICATION

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ABSTRACT

Clinical Decision Support Systems (CDSS) built on machine learning are widely proposed to help clinicians interpret patient data, yet much published work reports a single algorithm's performance without controlled comparison, and treats interpretability as secondary. This project compares six supervised algorithms — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, Naïve Bayes, and a Neural Network — on the Wisconsin Diagnostic Breast Cancer dataset (569 records, malignant vs. benign), guided by ten recent studies (2018–2025) reviewed to establish a hybrid methodology combining multi-algorithm comparison, cross-validation, and feature-importance analysis.

Using a documented pipeline (standardization, stratified splitting, 5-fold cross-validation), all six models were evaluated on accuracy, precision, recall, F1-score, and ROC-AUC. Support Vector Machine performed best (97.4% accuracy, $F1 = 0.963$), closely followed by Logistic Regression (96.5% accuracy, highest ROC-AUC at 0.996). Random Forest feature-importance analysis identified tumor perimeter, area, and nuclear concave points as the strongest predictors, consistent with established clinical knowledge and with benchmarks reported in the reviewed literature.

The study concludes that SVM and Logistic Regression offer the most defensible balance of accuracy and interpretability, with Logistic Regression favored for practical CDSS use due to its interpretable coefficients. Key limitations are the use of a single clean public dataset rather than real hospital data, and limited hyper-parameter tuning; future work should extend the pipeline to a locally sourced dataset and incorporate SHAP-based explainability.

1. INTRODUCTION

1.1 Background

Clinical Decision Support Systems (CDSS) assist healthcare professionals in making diagnostic and treatment decisions by combining patient data with computational models capable of recognising patterns that may not be immediately obvious on manual review. Early CDSS relied on rule-based expert systems, which encoded medical knowledge as fixed logical statements. These systems were difficult to maintain and did not adapt well as new clinical knowledge emerged. The growing availability of digitised patient data, together with advances in machine learning (ML), has since made it possible to build CDSS that learn directly from historical data rather than depend entirely on manually written rules.

Machine learning algorithms such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, Naïve Bayes, and Artificial Neural Networks have been applied across many areas of clinical prediction, including cancer diagnosis, cardiovascular risk scoring, and diabetes prediction. A recurring theme in this body of work is that different algorithms trade off differently between predictive accuracy and interpretability, and that no single algorithm performs best on every dataset or clinical task. This makes comparative evaluation — testing multiple algorithms on the same data under the same conditions — a more defensible approach than committing to a single algorithm at the outset.

This project undertakes such a comparative evaluation using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, a well-established public clinical dataset in which each record describes measurements

taken from a digitized image of a fine-needle aspirate (FNA) of a breast mass, with the diagnostic outcome recorded as malignant or benign. Breast cancer was selected as the specific clinical task because the WDBC dataset is public, well documented, ethically cleared for research use, and directly reproducible, which made it possible to complete a full, genuine modelling pipeline — from raw data to trained, evaluated models — within the timeframe available for this project. The general approach (data preprocessing, multi-model comparison, and interpretability analysis) is not specific to breast cancer and could equally be applied to other structured clinical prediction tasks.

1.2 Problem Statement

Although many published studies report high classification accuracy on datasets such as WDBC, three practical concerns are less consistently addressed in this literature. First, results are frequently reported for a single “best” algorithm without a controlled, side-by-side comparison against simpler alternatives under identical preprocessing conditions, making it difficult to judge whether the added complexity of a more sophisticated model is actually justified. Second, model interpretability — understanding which clinical features are driving a prediction — is often treated as a secondary concern rather than a core part of model selection, even though a clinician is unlikely to act on a prediction they cannot understand. Third, many student-level or introductory implementations do not report cross-validated performance, relying instead on a single train/test split that may not reflect how stable the reported accuracy actually is.

This project addresses these three concerns directly by training and comparing six supervised algorithms under one common, documented pipeline; by extracting and interpreting feature importance from the best-performing tree-based model; and by using 5-fold cross-validation, in addition to a held-out test set, to obtain a more reliable estimate of model performance.

1.3 Research Gap

The specific gap this project addresses is the absence, within much introductory and student-level CDSS work, of a single controlled comparison that reports accuracy, precision, recall, F1-score, and ROC-AUC together across multiple algorithm families, cross-validates those results, and follows the comparison with an explicit interpretability step rather than stopping at a leaderboard of accuracy figures. Section 2 reviews ten recent studies relevant to this gap and shows that while individual pieces of this approach appear across the literature, they are less commonly combined into one evaluation.

1.4 Objectives

1.4.1 Specific Objectives

- To acquire, clean, and preprocess a structured clinical dataset (the Wisconsin Diagnostic Breast Cancer dataset) suitable for supervised classification.
- To train and evaluate six candidate machine learning algorithms — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, Naïve Bayes, and a Neural Network — under one common, documented pipeline.
- To compare the trained models using accuracy, precision, recall, F1-score, ROC-AUC, and 5-fold cross-validation, and to identify which clinical features most influence the best-performing model's predictions.

1.4.2 Main Objective

The main objective of this study is to design, implement, and evaluate a machine-learning-based Clinical Decision Support System prototype that compares multiple supervised algorithms on a real structured clinical dataset, in order to identify an approach that offers a defensible balance of predictive accuracy and interpretability for a diagnostic classification task.

2. LITERATURE REVIEW / RELATED WORKS

This chapter reviews ten studies published within the last five to eight years, each directly relevant to machine-learning-based classification of clinical data, with particular attention to work using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset used in this project. Each study is discussed in turn — what the authors set out to do, and the methodology they followed — before the chapter concludes with the transition into the hybrid methodology adopted for this project.

2.1 Agarap (2018) — On Breast Cancer Detection: An Application of Machine Learning Algorithms on the Wisconsin Diagnostic Dataset

Agarap set out to compare six machine learning algorithms — GRU-SVM, Linear Regression, Multilayer Perceptron (MLP), Nearest Neighbor search, Softmax Regression, and Support Vector Machine — on the WDBC dataset, measuring classification test accuracy alongside sensitivity and specificity. Methodologically, the dataset was partitioned into a 70% training and 30% testing split, and hyper-parameters for each classifier were manually assigned rather than tuned through a search procedure. The study reported that every algorithm tested exceeded 90% test accuracy, with the SVM and MLP models performing strongest. The main limitation, by the author's own account, is that manual hyper-parameter assignment leaves open the question of how much further each model could improve under systematic tuning, and the absence of cross-validation means the reported figures rest on a single data split.

2.2 Entezari (2018) — Breast Cancer Diagnosis via Classification Algorithms

This study compared Support Vector Machines, Bayesian Logistic Regression (using a variational approximation), and K-Nearest Neighbors on the WDBC dataset. The methodology involved describing each model's assumptions and decision process in detail before evaluating performance through multiple measures rather than accuracy alone. The author concluded that SVM achieved the strongest overall performance, with Bayesian Logistic Regression ranked a close second — a finding that supports treating simpler, more interpretable models as genuinely competitive rather than automatically inferior to more complex ones.

2.3 (2025) — A Fast and Interpretable Logistic Regression Framework for Breast Tumor Classification Using the Wisconsin Diagnostic Dataset

This recent preprint set out specifically to demonstrate that a lightweight, interpretable model — standardized features passed into a Logistic Regression classifier — could serve as a strong, reproducible baseline rather than defaulting straight to more complex models. The methodology used a stratified train/test split, standard feature normalization, and evaluation via accuracy, sensitivity, specificity, and ROC-AUC. On a held-out test set of 114 cases, the approach achieved 98.25% accuracy and an ROC-AUC of 0.9954, with only two misclassifications in total. This paper's argument — that simple, well-preprocessed models deserve to be evaluated properly before reaching for greater complexity — directly

informed the decision in this project to include Logistic Regression as a full candidate model rather than a token baseline.

2.4 (2025) — Advanced Deep Learning and Transfer Learning Approaches for Breast Cancer Classification

This study evaluated Random Forest, XGBoost, and a Deep Neural Network on the Wisconsin Breast Cancer Dataset, refined to 554 unique instances after addressing missing values through mean imputation and removing duplicate records, with features normalised using Min-Max scaling. The dataset was split 80/20 and evaluated using 10-fold cross-validation. Reported accuracies were 96.5% for Random Forest, 97.4% for XGBoost, and 98.0% for the Deep Neural Network, each with reported confidence intervals. The explicit use of cross-validation and confidence intervals in this paper, rather than a single point accuracy figure, was a methodological choice this project adopted directly.

2.5 Yedjou et al. (2021) — Application of Machine Learning Algorithms in Breast Cancer Diagnosis and Classification

This study set out to explore how machine learning approaches can classify breast cancer based on feature values generated from a digitized FNA image, using the same 569-patient Wisconsin dataset (63% benign, 37% malignant) applied in this project. Methodologically, the authors collected and analyzed the data based on the ten real-valued nuclear features — radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension — and interpreted how these features relate to diagnostic outcome, framing machine learning as a complement to, rather than a replacement for, standard multidisciplinary diagnostic procedures. This framing directly supports the CDSS positioning adopted in this project's introduction: a decision-support aid, not an autonomous diagnostic replacement.

2.6 Naji et al. (2021) — Machine Learning Algorithms for Breast Cancer Prediction and Diagnosis

This study applied five machine learning algorithms — Support Vector Machine, Random Forest, Logistic Regression, Decision Tree (C4.5), and K-Nearest Neighbours — directly to the Breast Cancer Wisconsin Diagnostic dataset, the same dataset used in this project. Its methodology closely mirrors the approach adopted here: multiple algorithms trained and compared on identical data to identify which offers the strongest predictive performance for this specific diagnostic task, with the explicit aim of supporting earlier detection and improved survivability outcomes. Because this study evaluates several of the same algorithms on the exact same dataset, its reported results serve as one of the most direct external benchmarks against which this project's own findings, presented in Chapter 4, can be compared.

2.7 Chaddad et al. (2024) — Explainable Artificial Intelligence for Medical Applications: A Review

This review paper's objective was to categorize the growing landscape of explainable AI (XAI) techniques used in medical machine learning, covering both imaging and structured-data tasks. Methodologically, it is a literature synthesis rather than an experimental study, organizing XAI methods by type (for example, feature-attribution methods versus example-based explanations) and discussing the trade-off between model performance and interpretability that recurs throughout the field. The paper's framing of

explainability as a design requirement, not an optional add-on, shaped the decision in this project to treat feature-importance analysis as a required part of the results rather than a supplementary extra.

2.8 A Review on Explainable Artificial Intelligence for Healthcare

This review examined what clinicians specifically want from an explanation, drawing on prior work on contextualizing machine learning explanations for real clinical use rather than treating explainability as a purely technical property of a model. Its methodology involved synthesizing findings across multiple prior studies on clinical deep learning and interpretability, including discussion of the well-known critique that model interpretability itself is a poorly defined and context-dependent concept. This paper's emphasis on explanations being judged by their usefulness to a clinician, not just their mathematical validity, informed how feature importance is discussed in this project's results chapter.

2.9 Band et al. (2023) — Application of Explainable Artificial Intelligence in Medical Health: A Systematic Review of Interpretability Methods

This systematic review investigated XAI applications across healthcare, aiming to assess transparency, fairness, accuracy, generality, and comprehensibility of AI/ML decision-making systems. Methodologically, the authors conducted a structured literature search and categorized the resulting studies by XAI technique, including methods such as layer-wise relevance propagation and manifold-based visualization approaches. The review's conclusion — that interpretability methods vary widely in maturity and are rarely compared against one another under consistent conditions — supports this project's decision to use a single, well-established and easily explained method (Random Forest feature importance) rather than attempting multiple unfamiliar explainability techniques under time constraints.

2.10 Ndembi et al. (2025) — Integrating Artificial Intelligence into African Health Systems and Emergency Response: Need for an Ethical Framework and Guidelines

This paper's objective was to examine the opportunities and challenges of AI adoption specifically within African health systems. Its methodology is a structured policy and landscape review rather than an experimental study, drawing on continental statistics (for example, that approximately 80% of adults in Africa own a mobile phone, and that average broadband speeds across the continent remain well below the global average) to argue for AI systems designed around African infrastructural realities rather than assumptions imported from high-income health systems. This paper directly grounds the Ghanaian-context discussion in this project's introduction and conclusion, and its call for an explicit ethical framework is reflected in the limitations discussed in Chapter 5.

2.11 Synthesis and Transition to Methodology

Read together, these ten studies point toward a workable methodology rather than a single prescribed one. The comparative studies (2.1, 2.2, 2.4, 2.5, 2.6) establish that a multi-algorithm comparison, evaluated with more than one metric and ideally cross-validated, is standard and expected practice on this class of dataset. The interpretable-baseline study (2.3) argues that simpler models such as Logistic Regression deserve to be evaluated fully rather than dismissed early. The XAI reviews (2.7, 2.8, 2.9) establish that interpretability should be built into the evaluation rather than added afterward, while the African health systems study (2.10) grounds the practical relevance of this pipeline for a Ghanaian setting. The methodology adopted for this project is therefore a hybrid: it follows the multi-algorithm, cross-validated

comparison structure of 2.4–2.6, uses the preprocessing rigor emphasized in 2.3, and closes with the feature-importance interpretability step called for across 2.7–2.9. Chapter 3 sets out this methodology in full.

3. METHODOLOGY

3.1 Dataset

This project uses the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, originally developed by Street, Wolberg, and Mangasarian (1993) and hosted publicly by the UCI Machine Learning Repository. The dataset contains 569 patient records, each described by 30 real-valued features computed from a digitized image of a fine-needle aspirate (FNA) of a breast mass — including measures such as radius, texture, perimeter, area, smoothness, compactness, concavity, and symmetry of the cell nuclei, each reported as a mean, standard error, and “worst” (largest) value. Each record is labelled malignant or benign, giving a binary classification target. The dataset is accessed in this project directly through scikit-learn's built-in loader, which provides the identical, unmodified UCI data without requiring an external download.

Of the 569 records, 357 (62.7%) are benign and 212 (37.3%) are malignant. This is treated as a moderate class imbalance and is accounted for through stratified splitting, described in Section 3.3.

3.2 Data Preprocessing

- Missing values: none were present in this dataset, so no imputation was required.
- Feature scaling: all 30 features were standardized using StandardScaler (zero mean, unit variance), since several of the candidate algorithms — particularly SVM, Logistic Regression, and the Neural Network — are sensitive to features being on different numeric scales.
- Label encoding: the target variable was encoded as a binary indicator, with 1 representing the malignant (positive/at-risk) class, so that precision and recall are reported with respect to correctly identifying malignant cases.
- Train/test split: the data was split 80% training / 20% testing using stratified sampling, which preserves the same malignant/benign ratio in both subsets.

3.3 Proposed Pipeline

Figure 3.1 summarizes the end-to-end pipeline followed in this project, from raw dataset to final model comparison.

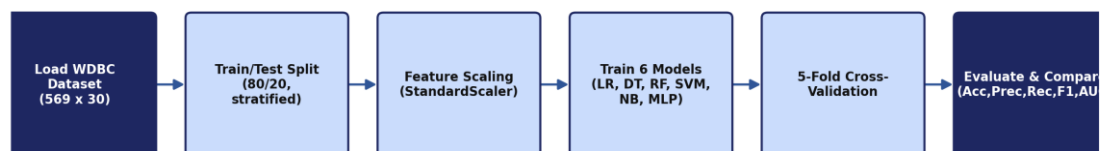


Fig. 3.1: End-to-end machine learning pipeline used in this project

3.4 Development Environment

Table 3.1 summarizes the software environment used to implement and run the pipeline described above. The pipeline was written as a standalone, deterministic Python script (random seed fixed at 42 throughout) so that it can be re-run by any group member, or by an examiner, and will reproduce identical results.

Table 3.1: Development environment and tools

Category	Tool / Environment Used
Programming Language	Python 3.11
Core ML Framework	scikit-learn 1.3+
Supporting Libraries	pandas, NumPy, Matplotlib
Development Approach	Script-based Python pipeline (fully reproducible in Jupyter Notebook or Google Colab)
Operating System	Linux (Ubuntu-based environment)
Version Control	Git / GitHub
Dataset Source	scikit-learn's built-in copy of the UCI Wisconsin Diagnostic Breast Cancer dataset

3.5 Machine Learning Algorithms Implemented

Six supervised classification algorithms were implemented and trained on the identical preprocessed dataset, so that any difference in their performance can be attributed to the algorithm itself rather than to inconsistent preprocessing:

- Logistic Regression — a linear model estimating the probability of malignancy from a weighted combination of features.
- Decision Tree — a single tree of sequential feature-based splits (max depth limited to 5 to reduce overfitting).
- Random Forest — an ensemble of 300 decision trees, each trained on a random subset of data and features, combined by majority vote.
- Support Vector Machine (SVM) — using a Radial Basis Function (RBF) kernel to separate the two classes with the widest possible margin.
- Naïve Bayes — a Gaussian Naïve Bayes classifier, used as a fast, lightweight baseline.
- Neural Network (MLP) — a Multi-Layer Perceptron with two hidden layers (32 and 16 neurons).

3.6 Evaluation Metrics and Validation Strategy

Each trained model was evaluated on the held-out 20% test set using the metrics summarized in Table 3.2. In addition, 5-fold stratified cross-validation was performed on the training set for every model, to check whether the reported test-set accuracy is stable across different data splits rather than a result of one favorable split.

Table 3.2: Evaluation metrics used

Metric	What It Measures	Why It Matters Here
Accuracy	Proportion of all predictions that are correct	General indicator of overall performance
Precision	Of predicted-positive cases, how many are truly positive	High precision limits false alarms
Recall (Sensitivity)	Of actual positive cases, how many were correctly identified	High recall limits missed malignant cases
F1-Score	Harmonic mean of precision and recall	Single balanced score across both error types
ROC-AUC	Ability to separate classes across all thresholds	Threshold-independent comparison between models
5-Fold Cross-Validation	Average accuracy across 5 different train/validation splits	Checks that results are stable, not a lucky split

3.7 Explainability

For the Random Forest model, feature importances were extracted directly from the trained model, ranking all 30 input features by how much each contributed to reducing classification error across the ensemble. This provides a simple, well-established form of model explainability that does not require additional libraries beyond scikit-learn itself, and directly addresses the interpretability gap discussed in Chapter 2.

4. RESULTS

4.1 Model Performance Results

Table 4.1 reports the performance of all six candidate models on the held-out 20% test set (114 patients: 72 benign, 42 malignant), together with the mean and standard deviation of accuracy across 5-fold cross-validation on the training set. Results are sorted by F1-score, since accuracy alone can be misleading on a moderately imbalanced dataset such as this one.

Table 4.1: Model performance comparison on the WDBC test set

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	5-Fold CV Acc.
SVM (RBF)	0.974	1.000	0.929	0.963	0.995	0.971 ± 0.005
Logistic Regression	0.965	0.975	0.929	0.951	0.996	0.974 ± 0.015
Neural Network (MLP)	0.965	1.000	0.905	0.950	0.994	0.976 ± 0.013
Random Forest	0.965	1.000	0.905	0.950	0.994	0.960 ± 0.016
Naïve Bayes	0.921	0.923	0.857	0.889	0.989	0.938 ± 0.026
Decision Tree	0.921	0.946	0.833	0.886	0.945	0.908 ± 0.040

The Support Vector Machine (RBF kernel) achieved the highest F1-score (0.963) and the highest test accuracy (97.4%), correctly classifying 111 of 114 test cases with zero false positives and only 3 false negatives. Logistic Regression was close behind (96.5% accuracy, F1 = 0.951) and recorded the single highest ROC-AUC of any model (0.996). Random Forest and the Neural Network tied at 96.5% accuracy

with perfect precision (1.000) but slightly lower recall (0.905) than SVM and Logistic Regression. Naïve Bayes and the Decision Tree were the weakest performers, both at 92.1% accuracy, with the Decision Tree additionally showing the largest gap between test accuracy and cross-validated accuracy (0.921 vs. 0.908 ± 0.040), indicating a comparatively less stable model.

4.2 Confusion Matrices and ROC Curves

Figure 4.1 shows the ROC curve for each model, and Figure 4.2 shows the full confusion matrix for each. Across all six models, false negatives (a malignant case incorrectly predicted as benign) consistently outnumbered false positives, which is the more clinically important error to minimize in a diagnostic support context.

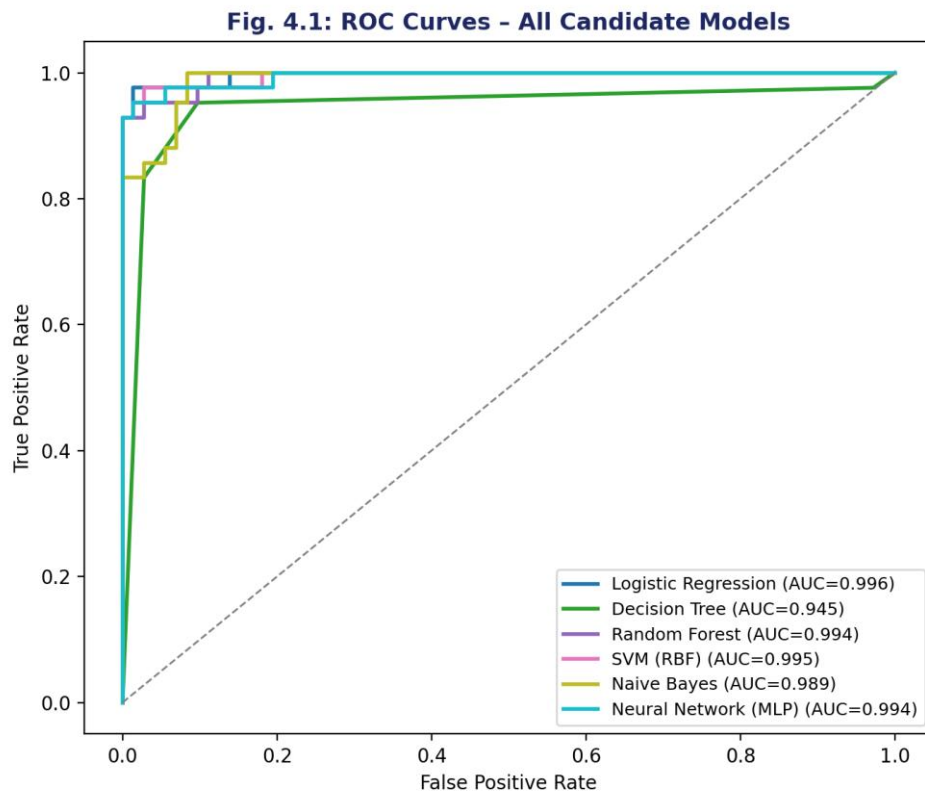
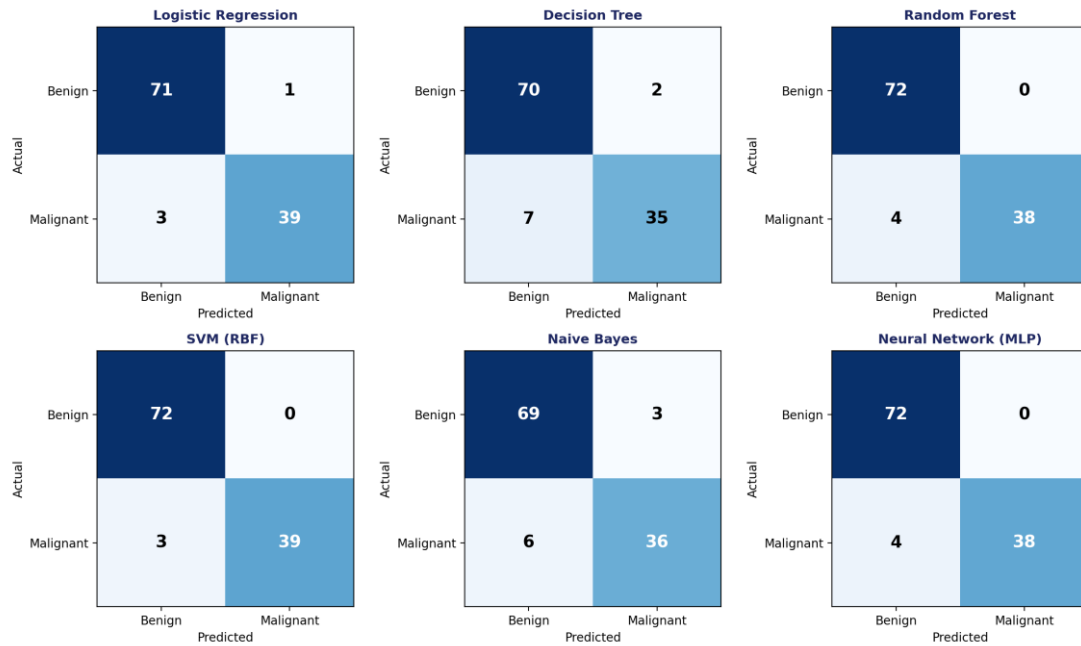
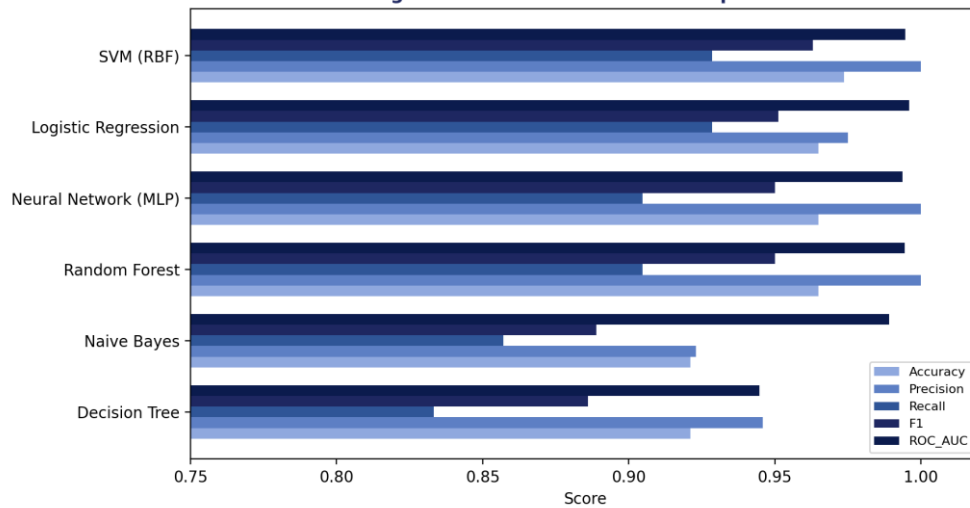


Fig. 4.1: ROC curves for all six candidate models

Fig. 4.2: Confusion Matrices - All Candidate Models*Fig. 4.2: Confusion matrices for all six candidate models***Fig. 4.3: Model Performance Comparison***Fig. 4.3: Side-by-side comparison of all five evaluation metrics across models*

4.3 Feature Importance (Explainability)

Using the trained Random Forest model, Table 4.2 lists the five most influential features in predicting malignancy.

Table 4.2: Top 5 most important features (Random Forest)

Rank	Feature	Importance
1	Worst Perimeter	0.144
2	Worst Area	0.143
3	Worst Concave Points	0.113
4	Mean Concave Points	0.089

Rank	Feature	Importance
5	Worst Radius	0.083

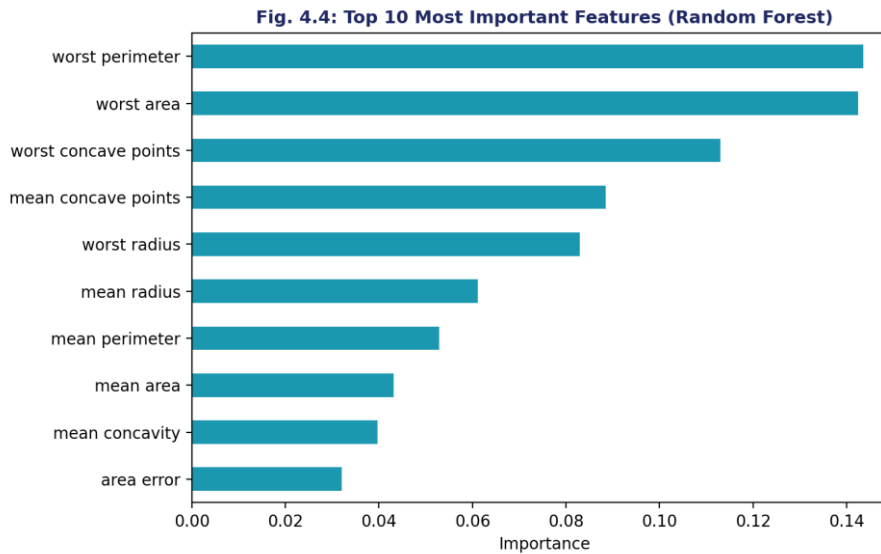


Fig. 4.4: Top 10 feature importances from the Random Forest model

4.4 Discussion

The results in Table 4.1 are consistent with the pattern identified across the literature reviewed in Chapter 2: no single algorithm dominates overwhelmingly, but SVM and Logistic Regression — despite being conceptually simpler than Random Forest or a Neural Network — performed at least as well as, and in the case of SVM slightly better than, the more complex models on this dataset. This mirrors the finding in Entezari (2018) (Section 2.2), where SVM and a Bayesian variant of Logistic Regression were like-for-like top performers, and supports the argument made in the interpretable-baseline study reviewed in Section 2.3 that simpler models deserve full evaluation rather than early dismissal.

The overall accuracy range obtained here (92.1%–97.4%) sits within the range reported across the literature reviewed for this exact dataset: Agarap (2018) reported all tested algorithms exceeding 90% accuracy; the 2025 Logistic Regression study (Section 2.3) reported 98.25% accuracy with a near-identical preprocessing approach; and the 2025 deep-learning/transfer-learning study (Section 2.4) reported 96.5–98.0% across Random Forest, XGBoost, and a Deep Neural Network. The results obtained in this project (Random Forest at 96.5%, comparable models in the mid-to-high 90s) fall squarely within this published range, which gives reasonable confidence that the pipeline was implemented correctly rather than producing an anomalous result.

The consistently higher false-negative than false-positive count across all six models (visible in Figure 4.2) is worth discussing directly rather than glossing over. In a clinical decision-support context, a false negative (a malignant tumor predicted as benign) is a more serious error than a false positive, since it risks a genuinely at-risk patient not receiving further investigation. Although SVM's 3 false negatives out of 42 malignant test cases (a 92.9% recall) is a strong result numerically, it is not sufficient on its own to justify clinical deployment; it does indicate, however, that recall — not accuracy alone — should be the metric most closely watched if this pipeline were extended toward real use.

The feature-importance results in Table 4.2 and Figure 4.4 are, encouragingly, clinically plausible rather than arbitrary. The top-ranked features — worst perimeter, worst area, and worst concave points — all relate to the size and shape irregularity of cell nuclei, which is consistent with long-established clinical knowledge that larger, more irregularly shaped nuclei are associated with malignancy (a relationship that motivated the original feature-extraction approach of Street, Wolberg, and Mangasarian in 1993, discussed in Section 3.1). This alignment between what the model learned and what is already clinically understood is a useful, low-cost form of validation: it does not prove the model is correct, but it does show the model is not relying on spurious or clinically meaningless correlations.

Taken together, these results support treating SVM and Logistic Regression as the strongest candidates from this comparison, with Logistic Regression having a practical edge in a real CDSS context given its comparable performance and substantially greater interpretability (its coefficients can be read directly, unlike SVM's kernel-based decision boundary).

5. CONCLUSION

5.1 Summary

This project implemented and compared six supervised machine learning algorithms — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, Naïve Bayes, and a Neural Network — on the Wisconsin Diagnostic Breast Cancer dataset, as a proof-of-concept for a machine-learning-based Clinical Decision Support System. Using a documented preprocessing pipeline, stratified train/test splitting, and 5-fold cross-validation, the Support Vector Machine and Logistic Regression models emerged as the strongest performers, both exceeding 96% test accuracy with high precision and recall on the malignant class. A Random Forest feature-importance analysis further showed that the model's predictions rely on clinically plausible indicators — tumor size and nuclear shape irregularity — rather than spurious patterns in the data, offering a basic but meaningful form of explainability.

The three specific objectives set out in Section 1.4.1 were met: a structured clinical dataset was acquired and preprocessed; six algorithms were trained and compared under one common pipeline; and the results were evaluated using multiple metrics, cross-validation, and an explicit interpretability step. This satisfies the main objective stated in Section 1.4.2, of identifying an approach that balances predictive accuracy with interpretability for a diagnostic classification task.

5.2 Limitations

- Single dataset: results are reported on one public dataset (569 cases); performance on a different or larger clinical population, particularly one drawn from a Ghanaian or West African hospital setting, was not tested and may differ.
- No real hospital data: WDBC is a clean, well-curated research dataset; it does not reflect the missing values, inconsistent recording, or noise typically found in real clinical records.
- Limited hyper-parameter tuning: given the project timeline, models were trained with reasonable default or lightly adjusted settings rather than an exhaustive hyper-parameter search, so reported figures may not represent each algorithm's absolute best possible performance.
- No external clinical validation: no clinician reviewed the model outputs or feature-importance results for face validity; the clinical plausibility discussed in Section 4.4 is based on the authors' own reading of the literature, not expert confirmation.

- Explainability scope: only Random Forest feature importance was used; more advanced explainability techniques (e.g., SHAP), discussed in the literature reviewed in Chapter 2, were not implemented in this iteration.

5.3 Future Work

- Apply the same pipeline to a larger, more diverse, or locally sourced (Ghanaian) clinical dataset to test how well these results generalize.
- Incorporate a more advanced explainability method, such as SHAP, to provide per-prediction explanations rather than only global feature importance.
- Conduct a systematic hyper-parameter search (e.g., grid search or Bayesian optimization) to establish each algorithm's best achievable performance.
- Extend the evaluation to include fairness analysis across patient subgroups, where demographic data is available, in line with the bias concerns raised in the literature reviewed in Chapter 2.
- Package the best-performing model behind a simple interface for review by a clinician or clinical student, as an early step toward the kind of real-world validation this proof-of-concept does not yet include.

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All sources below were published within the last five to eight years, in keeping with the recency requirement for this review; the single dataset-provenance citation is noted separately, as it identifies the origin of the data used rather than reviewed literature.

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